

J1 Using vectors

Name: _____

Class: _____

Date: _____

Time: **362 minutes**

Marks: **295 marks**

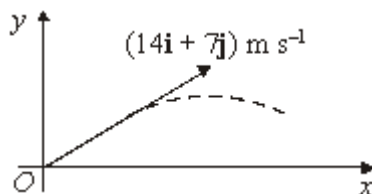
Comments:

EXAM PAPERS PRACTICE

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Q1.

Jonny is practising kicking a rugby ball. He kicks the ball from a point O on horizontal ground with velocity $14\mathbf{i} + 7\mathbf{j} \text{ m s}^{-1}$. The vectors $\mathbf{i}$ and $\mathbf{j}$ are in the directions of axes Ox and Oy , and Oy is vertically upwards.



- (a) (i) Find the coordinates of the ball t seconds after it has been kicked.

(3)

- (ii) Show that the equation of the path of the ball is

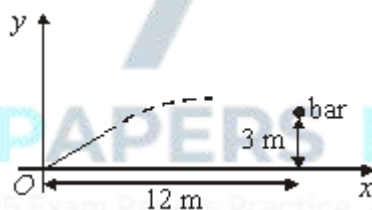
$$y = \frac{20x - x^2}{40}$$

(3)

- (iii) Find the horizontal range of the ball.

(2)

- (b) The ball moves towards a horizontal bar which is 3 metres above the ground and at a horizontal distance of 12 metres from O , as shown in the diagram.



Determine whether the ball passes above or below the bar.

(3)

- (c) State one modelling assumption that you have used in answering this question.

(1)

(Total 12 marks)

Q2.

Four forces, $\begin{bmatrix} 6 \\ 0 \end{bmatrix}$ newtons, $\begin{bmatrix} 0 \\ 4 \end{bmatrix}$ newtons, $\begin{bmatrix} -3 \\ -4.5 \end{bmatrix}$ newtons and $\begin{bmatrix} 3 \\ -2 \end{bmatrix}$ newtons, act on a particle.

- (a) Express the resultant, $\mathbf{F}$ newtons, of these four forces as a column vector.

(2)

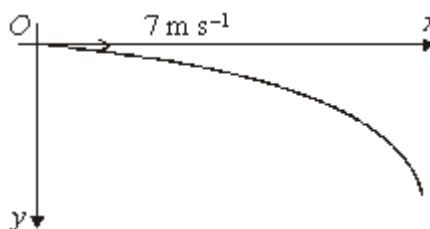
- (b) Find the magnitude of $\mathbf{F}$.

(2)

(Total 4 marks)

Q3.

Elaine kicks a ball off a cliff top. She kicks the ball from a point O and it subsequently moves in a vertical plane with respect to axes which are horizontal and vertically downwards, as shown in the diagram.



The initial velocity of the ball is horizontal and of magnitude 7 m s^{-1} .

- (a) Find the coordinates of the ball t seconds after it has been kicked.

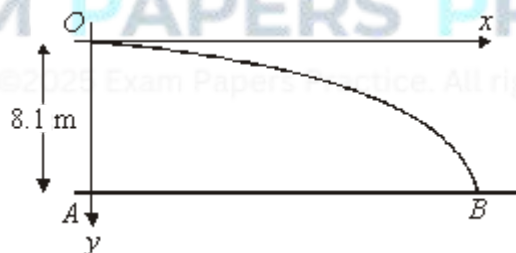
(3)

- (b) Show that the equation of the path of the ball is

$$y = \frac{x^2}{10}.$$

(2)

The ball subsequently lands on a horizontal beach. The point A on the beach is vertically below O and the ball lands at the point B on the beach, as shown in the diagram.



- (c) The distance OA is 8.1 metres.

Find the distance AB .

(2)

- (d) Find the speed of the ball as it reaches B .

(5)

(Total 12 marks)

Q4.

A particle P moves so that at time t seconds its position vector, $\mathbf{r}$ metres, is given by

$$\mathbf{r} = 6t\mathbf{i} + t^2\mathbf{j}.$$

- (a) Find the velocity of P at time t . (2)
- (b) Find an expression for the speed of P at time t . (2)
- (c) Find the value of t when P moves with speed $6\sqrt{2} \text{ m s}^{-1}$. (2)

(Total 6 marks)

Q5.

A particle P moves so that at time t it has velocity

$$\mathbf{v} = 4\mathbf{i} - 2t\mathbf{j}, \quad t \geq 0.$$

When $t = 0$, the particle P has position vector $8\mathbf{j}$.

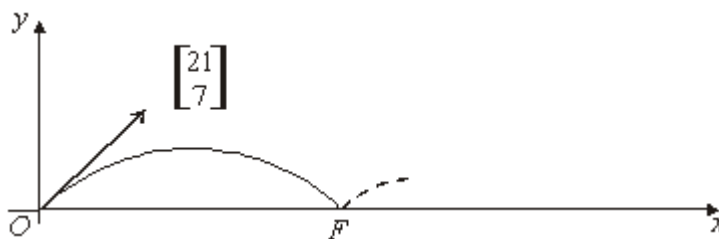
- (a) Find the position vector of P at time t . (4)
- (b) Find the position vector of P when $t = 2$. (1)
- (c) Find the time when the position vector of P is in the direction of the vector $\mathbf{i}$. (2)

(Total 7 marks)

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Q6.

A ball is kicked from a point O on horizontal ground with velocity $\begin{bmatrix} 21 \\ 7 \end{bmatrix} \text{ m s}^{-1}$.
The ball subsequently hits the ground at a point F , as shown in the diagram.



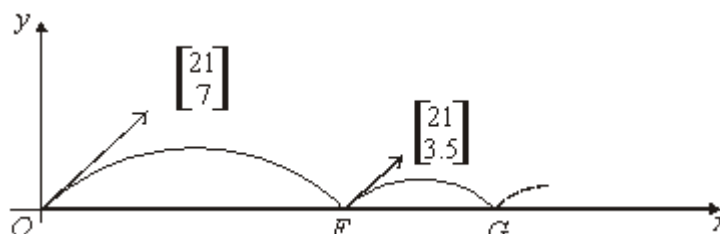
- (a) (i) Find the time between the ball being kicked and reaching F . (3)

(ii) Find the distance OF .

(2)

- (b) The ball rebounds from F with velocity $\begin{bmatrix} 21 \\ 3.5 \end{bmatrix}$ m s⁻¹. It subsequently hits the ground again at

G and then rebounds, as shown in the diagram below.



Find the distance FG .

(4)

- (c) After bouncing at G , the ball rebounds with velocity $\begin{bmatrix} 21 \\ 1.75 \end{bmatrix}$ m s⁻¹, so that the vertical component of the velocity is halved again, and then subsequently bounces next at H .
Find the total distance OH .

(3)

(Total 12 marks)

Q7.

A particle P moves so that at time t seconds its velocity $\mathbf{v}$ m s⁻¹ is

$$\mathbf{v} = 2t\mathbf{i} + 4\mathbf{j}, \quad t \geq 0.$$

- (a) At time $t = 0$, the particle is at the point with position vector $2\mathbf{j}$ metres.

Find the position vector of P at time t .

(4)

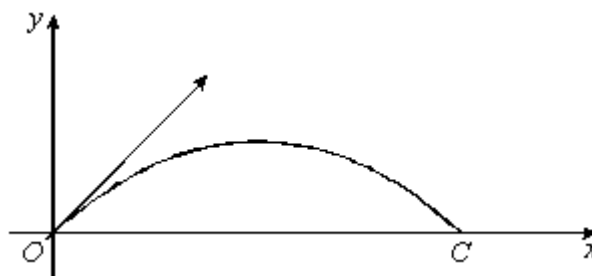
- (b) At times $t = 2$ and $t = 4$, the particle passes through the points C and D respectively.

Find the vector $\overrightarrow{CD}$.

(4)

(Total 8 marks)

Q8.



Oliver throws a ball from point O , with velocity $\begin{bmatrix} 7 \\ 7 \end{bmatrix}$ m s⁻¹. The ball is subsequently caught by

Chris at a point C which is on the same horizontal level as O , as shown in the diagram. During the motion from O to C the ball moves in a vertical plane.

- (a) Taking O as the origin, find the position vector of the ball t seconds after being thrown.

(3)

- (b) Show that the equation of the path of the ball is

$$y = x - 0.1x^2,$$

where (x, y) are the coordinates at time t of the ball relative to the origin O .

(4)

- (c) A horizontal wire, which is part of a fence, cuts the plane of motion of the ball at the point $(4, 2)$.

Determine whether the ball passes above or below the wire.

(3)

- (d) Find the distance OC .

(4)

- (e) Find the greatest speed of the ball during its motion from O to C .

(2)

(Total 16 marks)

Q9.

At time t , the position vector of a particle Q is

$$\mathbf{r} = (t^2 - 6t + 4)\mathbf{i} + \frac{1}{3}t^3\mathbf{j}.$$

- (a) Find the velocity of Q at time t .

(3)

- (b) Find the value of t when Q is moving parallel to the vector $\mathbf{j}$.

(2)

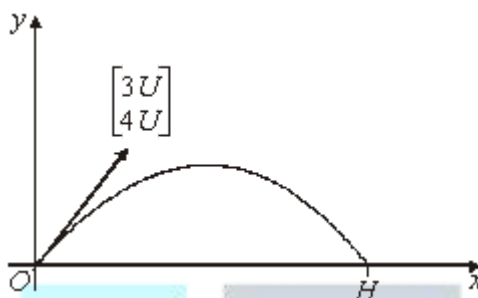
- (c) Find the acceleration of Q and state, with a reason, whether or not this acceleration is constant.

(2)

(Total 7 marks)

Q10.

George throws a ball from a point O , with velocity $\begin{bmatrix} 3U \\ 4U \end{bmatrix}$. The ball subsequently lands at a point H , which is at the same horizontal level as O , as shown in the diagram.



- (a) Show that the time taken by the ball to travel from O to H is $\frac{8U}{g}$. (3)
- (b) Find, in terms of g and U , the distance OH . (2)
- (c) Find, in terms of U , the initial speed of the ball. (2)
- (d) Find, in terms of g and U , the two times during the flight from O to H when the ball is moving with speed $\sqrt{18U}$. (6)

(Total 13 marks)

Q11.

A particle moves with constant acceleration in a horizontal plane. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular and lie in the plane. At time $t = 0$, the particle is at rest at the origin. At time $t = 4$ seconds, the position vector of the particle is $(-16\mathbf{i} + 16\mathbf{j})$ metres.

- (a) Show that the acceleration of the particle is $(-2\mathbf{i} + 2\mathbf{j}) \text{ m s}^{-2}$. (4)
- (b) Find the speed of the particle when $t = 5$.

(4)

- (c) The mass of the particle is 3 kg. Two horizontal forces **P** and **Q** act on the particle. If $\mathbf{P} = 20\mathbf{i} - 10\mathbf{j}$, find **Q**.

(4)

(Total 12 marks)

Q12.

A particle moves so that, at time t seconds, its position vector, $\mathbf{r}$ metres, is given by

$$\mathbf{r} = 4e^{-t}\mathbf{i} + (6t + 3e^{-t})\mathbf{j}$$

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular.

- (a) Show that the velocity of the particle is $(-4\mathbf{i} + 3\mathbf{j}) \text{ m s}^{-1}$ when $t = 0$. (3)
- (b) Find an expression for the acceleration of the particle at time t . (2)
- (c) Find the magnitude of the acceleration of the particle when $t = 0$. (2)
- (d) Describe what happens to the velocity of the particle as t becomes large. (2)

(Total 9 marks)

Q13.

A particle moves with constant acceleration $(4\mathbf{i} + 2\mathbf{j}) \text{ m s}^{-2}$. At time $t = 0$, the particle is at the origin and has velocity $(3\mathbf{i} - 10\mathbf{j}) \text{ m s}^{-1}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Show that the velocity of the particle at time t seconds is $(3 + 4t)\mathbf{i} + (2t - 10)\mathbf{j}$. (2)
- (b) Find the time when the particle is travelling due east. (2)
- (c) (i) Find the position vector of the particle when $t = 10$. (3)
- (ii) Find the velocity of the particle when $t = 10$. (1)
- (d) The particle stops accelerating when $t = 10$ and then moves with constant velocity. Find the distance of the particle from the origin after it has been travelling with this

constant velocity for a further 10 seconds.

(6)

(Total 14 marks)

Q14.

A particle is initially at the origin and has velocity $(4\mathbf{i} + 3\mathbf{j}) \text{ m s}^{-1}$. It moves with constant acceleration, and 10 seconds later has velocity $(5\mathbf{i} - 2\mathbf{j}) \text{ m s}^{-1}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Show that the acceleration of the particle is $(0.1\mathbf{i} - 0.5\mathbf{j}) \text{ m s}^{-2}$.

(3)

- (b) Find an expression for the position vector of the particle at time t seconds.

(2)

- (c) Find the time when the particle is due east of the origin.

(4)

- (d) Find the time when the particle is moving due east.

(4)

(Total 13 marks)

Q15.

At time $t = 0$, a boat has velocity $(2\mathbf{i} - 3\mathbf{j}) \text{ m s}^{-1}$ and is at the origin. After moving with a constant acceleration of $\mathbf{a} \text{ m s}^{-2}$ for 20 seconds, its position vector is $(60\mathbf{i} + 20\mathbf{j})$ metres. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find $\mathbf{a}$.

(3)

- (b) Find the velocity of the boat at time t seconds.

(2)

- (c) Find the speed of the boat when it is travelling south east.

(5)

(Total 10 marks)

Q16.

A particle is initially at rest at the point A with position vector $(9\mathbf{i} + 10\mathbf{j})$ metres, with respect to an origin O . It moves with constant acceleration, and 10 seconds later is at the point B with position vector $(19\mathbf{i} - 25\mathbf{j})$ metres. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular.

- (a) Show that the acceleration of the particle is $(0.2\mathbf{i} - 0.7\mathbf{j}) \text{ m s}^{-2}$.

(4)

- (b) Find the speed of the particle at B .

(4)

- (c) Verify that the particle passes through the point with position vector $(15.4\mathbf{i} - 12.4\mathbf{j})$ m.

(4)

- (d) Draw a diagram to show the path of the particle as it moves from A to B . You should include the unit vectors $\mathbf{i}$ and $\mathbf{j}$ on your diagram.

(2)

(Total 14 marks)

Q17.

A particle moves in a horizontal plane. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular and lie in the horizontal plane.

Initially the particle is at the origin and has velocity $(5\mathbf{i} - 2\mathbf{j})$ m s⁻¹. It moves with constant acceleration. At time $t = 10$ it has position vector $(40\mathbf{i} - 15\mathbf{j})$ metres.

- (a) Show that the velocity of the particle is $(3\mathbf{i} - \mathbf{j})$ m s⁻¹ when $t = 10$.

(4)

- (b) Find the acceleration of the particle.

(3)

- (c) The mass of the particle is 15 kg. Find the magnitude of the resultant force acting on the particle.

(4)

(Total 11 marks)

Q18.

A particle of mass 4 kg moves on a smooth horizontal plane. It is initially at rest at the origin. A force $\mathbf{F} = (8\mathbf{i} - 12\mathbf{j})$ N acts on the particle for 20 seconds. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular and lie in the horizontal plane.

- (a) (i) Find the acceleration of the particle.

(2)

- (ii) Find the velocity of the particle at the end of the 20 second period.

(2)

- (iii) Find the position of the particle at the end of the 20 second period.

(3)

- (b) At the end of the 20 second period, the force $\mathbf{F}$ is removed. Find the distance of the particle from the origin after the particle has been in motion for a total of 45 seconds.

(5)

(Total 12 marks)

Q19.

A boat moves so that its position vector, $\mathbf{r}$ metres, at time t seconds is given by

$$\mathbf{r} = (4t - 0.01t^2)\mathbf{i} + (5 - 3t - 0.04t^2)\mathbf{j}$$

where the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) State the position of the boat when $t = 0$. (1)
 - (b) Find the time when the boat is due south of its initial position found in part (a). (4)
 - (c) Find the time when the boat is travelling south east. (5)
- (Total 10 marks)

Q20.

A model aeroplane moves in a horizontal plane with a constant acceleration. Initially the aeroplane is at the origin and has velocity $(35\mathbf{i} + 45\mathbf{j}) \text{ m s}^{-1}$. After accelerating for 8 seconds, the velocity of the aeroplane is $(19\mathbf{i} + 13\mathbf{j}) \text{ m s}^{-1}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular and lie in the horizontal plane.

- (a) Show that the acceleration of the aeroplane is $(-2\mathbf{i} - 4\mathbf{j}) \text{ m s}^{-2}$. (3)
 - (b) Find an expression for the position vector of the aeroplane at time t seconds. (3)
 - (c) Find the time when the position vector of the aeroplane is $(300\mathbf{i} + 225\mathbf{j}) \text{ m}$. (7)
- (Total 13 marks)

Q21.

Two forces $\mathbf{F}_1 = (3\mathbf{i} + 4\mathbf{j})\text{N}$ and $\mathbf{F}_2 = (6\mathbf{i} - 8\mathbf{j})\text{N}$, act on a particle. The resultant of these two forces is $\mathbf{F}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular.

- (a) Find $\mathbf{F}$. (2)
 - (b) Find the magnitude of $\mathbf{F}$. (2)
 - (c) Find the acute angle between $\mathbf{F}$ and the unit vector $\mathbf{i}$. (3)
- (Total 7 marks)

Q22.

A particle is initially at the origin. The particle has a constant acceleration of $(2\mathbf{i} - 3\mathbf{j}) \text{ m s}^{-2}$. Two seconds after it has left the origin the velocity of the particle is $(2\mathbf{i} + 14\mathbf{j}) \text{ m s}^{-1}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular.

- (a) Show that the initial velocity of the particle is $(-2\mathbf{i} + 20\mathbf{j}) \text{ m s}^{-1}$. (3)
- (b) Find the distance of the particle from the origin after 3 seconds. (5)
- (c) The speed of the particle is 10 m s^{-1} for the first time T seconds after it has left the origin. Find T . (6)
- (Total 14 marks)

Q23.

A particle has mass 2000 kg. A single force, $\mathbf{F} = 1000t\mathbf{i} - 5000t\mathbf{j}$ newtons, acts on the particle, at time t seconds. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular. No other forces act on the particle.

- (a) Find an expression for the acceleration of the particle. (2)
- (b) At time $t = 0$, the velocity of the particle is $6\mathbf{j} \text{ m s}^{-1}$. Show that at time t the velocity, $\mathbf{v} \text{ m s}^{-1}$, of the particle is given by
- $$\mathbf{v} = \frac{t^2}{4}\mathbf{i} + \left(6 - \frac{5t}{2}\right)\mathbf{j}$$
- (4)
- (c) The particle is initially at the origin. Find an expression for the position vector, $\mathbf{r}$ metres, of the particle at time t seconds. (4)
- (Total 10 marks)

Q24.

A particle moves with a constant acceleration of $(-2\mathbf{i} - 4\mathbf{j}) \text{ m s}^{-2}$. At time $t = 0$ the particle is at the origin and has velocity $(4\mathbf{i} + 7\mathbf{j}) \text{ m s}^{-1}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular.

- (a) Find an expression for the position vector of the particle at time t seconds. (2)
- (b) Find the time when the particle has position vector $(3\mathbf{i} + 5\mathbf{j})$ metres. (6)
- (Total 8 marks)

Q25.

A boat moves with a constant acceleration of $(0.2\mathbf{i} + 0.1\mathbf{j}) \text{ m s}^{-2}$, where the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively. Time t is measured in seconds.

At time $t = 10$, the boat is at the point A , which has position vector $(30\mathbf{i} - 35\mathbf{j})$ metres.

At time $t = 20$, the boat is at the point B , which has position vector $(70\mathbf{i} - 40\mathbf{j})$ metres.

- (a) Find the vector $\overrightarrow{AB}$. (1)
 - (b) Find the velocity of the boat when $t = 10$. (3)
 - (c) Using your answer to part (b), find the velocity of the boat when $t = 0$. (3)
 - (d) Find the position vector of the boat when $t = 0$. (3)
- (Total 10 marks)**

Q26.

A boat moves so that its position vector, $\mathbf{r}$ metres, at time t seconds is

$$\mathbf{r} = 40 \cos\left(\frac{t}{20}\right) \mathbf{i} + 80 \sin\left(\frac{t}{20}\right) \mathbf{j}$$

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find an expression for the velocity of the boat at time t . (3)
 - (b) In what direction is the boat travelling when $t = 0$? Justify your answer. (2)
 - (c) At what time is the boat travelling due south for the first time? (2)
- (Total 7 marks)**

Q27.

A particle moves on a horizontal plane subject to the following three forces:

$$\mathbf{F}_1 = (2\mathbf{i} + 3\mathbf{j})\text{N}, \quad \mathbf{F}_2 = (6\mathbf{i} - 8\mathbf{j})\text{N} \quad \text{and} \quad \mathbf{F}_3 = (-2\mathbf{i} + 10\mathbf{j})\text{N}$$

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular and lie in the horizontal plane.

Do not include any other forces in your calculations.

- (a) Find the resultant force on the particle. (2)
- (b) The mass of the particle is 8 kg. Find the acceleration of the particle. (1)
- (c) At time $t = 0$, the particle is at the origin and has velocity $(3\mathbf{i} - 2\mathbf{j})\text{ms}^{-1}$.

- (i) Find the velocity of the particle when $t = 40$ seconds. (2)
- (ii) Find the distance of the particle from the origin when $t = 40$ seconds. (5)
- (Total 10 marks)

Q28.

At time $t = 0$, a particle is at the origin and moving with velocity $(4\mathbf{i} + 2\mathbf{j}) \text{ m s}^{-1}$.
When $t = 10$ seconds the position vector of the particle is $(44\mathbf{i} + 28\mathbf{j})$ metres.
The particle is moving with constant acceleration.

- (a) Find the acceleration of the particle. (4)
- (b) Find the position vector of the particle at time t . (3)
- (c) At time $t = T$, the position vector of the particle is $(96\mathbf{i} + 72\mathbf{j})$ metres.
- (i) Find T . (4)
- (ii) Find the speed of the particle at this time. (3)

(Total 14 marks)